

Macroinvertebrate diversity and relationship with environmental variables in the headwater streams of Toebirongchhu sub-watershed, Bhutan

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ABSTRACT

Macroinvertebrates are important components of biodiversity in headwater streams, and their diversity is driven by various environmental variables. The study was carried out in Toebirongchhu sub-watershed to study: (i) the macroinvertebrate diversity and composition within whole study area, three streams and the reaches, and (ii) relationship between macroinvertebrate diversity and environmental variables. The study area had low diversity considering Shannon-Weiner diversity ($H' = 1.69$), but high while considering Simpson diversity ($D' = .28$), and a total of 829 individuals, 18 families and one class was recorded. Toebirongchhu stream had the highest H' (1.71) and Jichulum the lowest D' (.07) and the highest taxon richness ($S = 16$), while reach T1 had the highest H' (1.72), lowest D' (.18), and the highest S (8) along with J1. Trichoptera (82.03%) the most and Plecoptera (0.24%) the least abundant order, while Hydropsychidae (37.0%) the most, and Athericidae and Polymitarciidae (0.1%) the least abundant families. There was no significant variations in environmental variables among streams and reaches, and pH was slightly alkaline (7.67-7.97) and temperature ranged from 8-11.6°C, but conductivity was higher in Jichulum ($J1 = 154.8 \mu S \text{ cm}^{-1}$; $J3 = 150.7 \mu S \text{ cm}^{-1}$). Stream width had significant correlation with H' ($\tau = .511$, $p = .04$), while conductivity had the highest correlation coefficient with S ($\tau = .65$, $p = .08$) and high correlation with H' and J' ($\tau = -.48$, $p = .19$). Among diversity indices S had stronger correlation with the maximum environmental variables showing it to be better diversity indicator in the study area.

KEYWORDS

Macroinvertebrates, diversity, freshwater, environmental variables, Toebirongchhu sub-watershed

Introduction

Macroinvertebrates are important components of biodiversity in headwater stream ecosystems, and headwater streams harbour macroinvertebrates that are headwater specialist, undescribed species and provide high β -diversity (Clarke *et al.*, 2008). Also headwater streams are prominent features in the landscape and are important sources of water, sediments and biota for downstream reaches (Clarke *et al.*, 2008). But the provision of freshwater for consumption and irrigation by headwaters had taken precedence over other commodities and services provided by freshwater ecosystems (Baron *et al.*, 2002) in the past, while by 1970s ecosystem health for itself has become valuable (Bonada *et al.*, 2006), and hence deterioration of aquatic biodiversity has become main concern in recent conservational and ecological studies in various parts of the world (Che Salmah *et al.*, 2013a).

In Bhutan too the rivers, streams, lakes and wetlands are viewed as crucial components of freshwater ecosystems for biodiversity and sustainable economic growth (Wangchuck Centennial Park (WCP), University of Calcutta (UoC) & WWF Bhutan, 2012), but very few studies have been done on the freshwater biodiversity to either quantify it or assess quality of freshwater for human use or for ecological integrity of freshwater (Giri & Singh, 2012, 2013; Dorji, Thinley & Jamtsho, 2014). Rapid assessment of Punatsangchhu basin found majority of sub-watersheds pristine and normal, and only few critical, but many sub-watersheds were found critical by function including the Toebirongchhu sub-watershed (Wangchuk, 2011). But the

study did not take into consideration the freshwater organisms, and already change in macroinvertebrate diversity and composition due to water quality deterioration caused by landuse change has been recorded in Bhutan (Giri & Singh, 2013; Dorji *et al.*, 2014). The knowledge of macroinvertebrate diversity is very limited in Hindu Kush Himalayan region (Stubauer *et al.*, 2010). In Bhutan the study of macroinvertebrate diversity *per se* has been very limited, for instance study was done within Wangchuk Centennial Park by WCD *et al.* (2012) and in two streams in Thimphu and Trongsa districts by Dorji *et al.* (2014). It is very important to study macroinvertebrate diversity in freshwater streams of Bhutan and particularly in Toebirongchhu sub-watershed as it is critical by function (Wangchuk, 2012) in order to record macroinvertebrate diversity and composition, and its relationship with environmental factors. Macroinvertebrate diversity in the streams are driven by various environmental variables like sediment (Jones *et al.*, 2012), discharge (Benson *et al.*, 2013), floods (Robinson, 2012), stream velocity, substrate structure and water quality (Che Salmah *et al.*, 2013b,), riparian-reach variable (Rios & Bailey, 2006), habitat structure (Barnes *et al.*, 2013), stream order (Clarke *et al.*, 2008), latitude (Mori *et al.*, 2010) and historic processes (Rader *et al.*, 2012). Besides, impact of anthropogenic activities like catchment landuse and nutrient concentrations (Weijters *et al.*, 2009), irrigation water withdrawal from the streams (Miller *et al.*, 2007), forest fragmentation (Che Salmah *et al.*, 2013a), road-to-stream connections (Miller & Miller, 2005), and water pollution

from sewage discharge (Canobbio, 2009) also impact macroinvertebrate diversity. Therefore, the objectivities of the current study were to: (i) study macroinvertebrate diversity and composition within the study area, streams and reaches, and (ii) relate the macroinvertebrate diversity with different environmental variables in the study area.

Materials and Methods

Study Area

The current study area falls within Toebirongchhu sub-watershed in Punakha district, in Western Bhutan. Two first order streams, Jichulum and Yuelghona, about 8 km and 23 km respectively from Messina town towards Thimphu, and lower portion of Toebirongchhu about 3 km from Messina town towards Punakha were selected for the study. Jichulum and Yuelghona streams are both tributaries of Toebirongchhu, while Toebirongchhu is a 3rd order stream that joins Punatsangchhu river (Fig. 1).



Figure 1. Toebirongchhu sub-watershed with named study streams. Inset Bhutan map with district boundaries (black), major rivers (green) and Punakha district (light blue).

Sampling Sites

In Jichulum and Yuelghona streams four sampling reaches of 20m each, two above and two below the Thimphu-Punakha highway were selected by walking about 50-100 m below and above the road to locate the first two sampling reaches, and other two were located by walking about 100 m farther away from the road, while in lower portion of Toebirongchhu one reach each was selected above and below Thimphu-Punakha highway. The coordinates and altitude for the reaches along with reach notations are given in Table 1.

Table 1. Streams with reach notation, latitude, longitude and altitude.

Stream name	Reach	Latitude	Longitude	Altitude(m)
Yuelghona	Y1	27° 31'03.06"	89° 48'26.56"	2326
	Y2	27° 31'02.37"	89° 45'24.80"	2334
	Y3	27° 31'03.07"	89° 45'18.59"	2337
	Y4	27° 31'02.85"	89° 45'16.07"	2359
Jichulum	J1	27° 31'12.59"	89° 49'57.54"	1598
	J2	27° 31'22.09"	89° 49'59.99"	1579
	J3	27° 31'08.07"	89° 49'55.51"	1627
	J4	27° 30'59.53"	89° 49'48.35"	1680
Toebirongchhu	T1	27° 31'30.48"	89° 52'05.47"	1254
	T2	27°31'35.99"	89° 51'57.93"	1279

Sampling environmental variables

The environmental variables grouped as geographical elements, hydrological elements and physicochemical

elements modified after Li *et al.*, (2012) were recorded into the printed field data sheet. Geographical elements included latitude, longitude and altitude which were determined using handheld Garmin GPS; hydrological elements for which average of three readings were recorded include stream width and depth that were measured using measuring tape, and stream velocity measured by timing a floating object over a stretch of 10m (Soldner *et al.*, 2004). The physicochemical elements were water temperature (measured using thermometer in the sites), pH and conductivity (assessed in the laboratory within 12 hours from the time of the water samples collection in the field using Hanna Bench-top pH and conductivity meters).

Sampling macroinvertebrates

The macroinvertebrates as well the environmental variables were sampled in different reaches of the streams from December 2012 to February 2013. In each reach three kick sampling of macroinvertebrates using D-frame nets (30 cm x 30 cm area, 500 µm mesh size) were done in riffles if it was the dominating in-stream habitat. The area of about 30 cm x 30 cm in front of the net was stirred for 2-3 minutes and then the net lifted with a forward scooping motion to capture all the macroinvertebrates floating in front of the net. If there were higher percentage of other in-stream habitats then one sampling was done in those habitats and other two in the riffle. The sampled macroinvertebrates in each reach were segregated from the debris and stored in 10% formalin for identification in the laboratory. The samples from each sampling unit within the reach were pooled for analysis. The identification was done till family level, except Oligochaeta, for appropriate analysis and also due to limited taxonomical works on macroinvertebrates in the country. The literatures used for the identification are Thorp & Rogers (2011); Galuti (2012); Gerber & Gabriel (2002); Hartmann (2007) and Nesemann *et al.*, (2011).

Statistical analysis

The diversity of macroinvertebrate at each reach, different streams and for the whole study area was computed using Shannon-Weiner index (H'), Simpson index (D'), taxon richness as total number of taxa (S), Shannon evenness ($J = H'/\ln S$) (Ntongani & Andrew, 2013), and relative abundance as percentage of total individuals belonging to a particular taxon out of total number of individuals within the reach, stream or whole study area respectively for the reach, stream and whole study area. In order to find the correlation between macroinvertebrate diversity and different environmental variables partial correlation (Kendall's tau) was performed using SPSS 16.0 to find specific correlation between environmental factors and different diversity indices as the raw data upon analysis was not normally distributed and the data set was limited (Field, 2009).

Result and Discussion

Macroinvertebrate diversity and abundance

Total of 829 individuals of macroinvertebrates were sampled from the whole study area and was comprised of 10 orders, 18 families and class Oligochaeta (Table 2). The

macroinvertebrate diversity was low in the study area as reflected by Shannon-Weiner diversity index ($H' = 1.67$) (Table 3) which ranges between 1 and 6 for natural communities (Stiling, 1996 in Flores & Zafaralla, 2012). It could be due to low evenness ($J' = .06$) as H' is affected by J' , but considering D' diversity was high ($D' = .28$).

Table 2. Macroinvertebrates in the study area with order, family, number of individuals, and relative abundance of family and order.

Order	Families	No. of individuals	Relative abundance (Family)	Relative abundance (Order)
Trichoptera	Hydropsychidae	306	37.0	
	Lepidostomatidae	297	35.9	82.03
	Polycentropodidae	77	9.3	
Ephemeroptera	Heptageniidae	34	4.1	
	Ephemerellidae	9	1.1	
	Baetiscidae	4	0.5	7.72
	Baetidae	16	1.9	
	Polymitarcyidae	1	0.1	
Plecoptera	Perlidae	2	0.2	0.24
Odonata	Libellulidae	7	0.8	3.86
Diptera	Euphaeidae	25	3.0	
	Athericidae	1	0.1	0.36
	Culicidae	2	0.2	
Megaloptera	Corydalidae	8	1.0	0.97
Lepidoptera	Pyralidae	11	1.3	1.33
Coleoptera	Gyrinidae	13	1.6	1.57
Tricladida	Planariidae	4	0.5	0.48
Hydracarina	Hydrachnellae	4	0.5	0.48
Oligochaeta*	Oligochaeta*	8	1.0	0.97
11	19	829	100	100

* Taxonomic level Class.

Among the streams also the diversity was relatively less with Toebirongchhu having the highest H' (1.71) and correspondingly evenness (0.16), Yuelghona in between and Jichulum the lowest, while Jichulum had the higher diversity when Simpson index (0.07) and the taxon richness ($n=16$) were considered (Table 3). The minor differences in the indices within the study could be due to their difference in sensitivity to abundant and rare taxa (Boyle & Fraleigh Jr., 2003) as H' and D' showed significant negative correlation overall as expected since diversity increases with decrease in D' and increase in H' ($\tau = -.69$, $P < .001$). The reaches also showed lower diversity corresponding to whole study area and the streams (Table 5). The reach T1 and the Toebirongchhu stream had the highest diversity among the streams and reaches possibly due to the higher stream order (3rd order) over other two (1st order) as macroinvertebrate diversity increases with increase in stream order from the lower to mid order (Clarke *et al.*, 2008; Flores & Zafaralla, 2012; Che Salmah *et al.*, 2013b). It may be also due to other factors like temperature, depth (Flores & Zafaralla, 2012) and velocity (Che Salmah *et al.*, 2013b) as these factors are significantly correlated with stream order in the study area and are correlated with diversity (Flores & Zafaralla, 2012). The low diversity in the study area, streams or reaches could be natural since headwater streams have lower α diversity (Clarke *et al.*, 2008) and the abundance and richness of the study area is similar to the results of studies done in other parts of Bhutan with taxonomic resolution at order levels (WCP, 2012 & Dorji *et al.*, 2014). It could be low due to anthropogenic impact as diversity decreases with human impact (Che Salmah *et al.*, 2013b, Giri & Singh, 2013), and human impacts like roadside drainage discharging into all

the three streams, especially above reaches Y1, Y2, J1, J2, T1 and T2, cattle grazing in almost all the reaches, and agricultural practice nearby T1, T2, J3 and J4 were observed. But, it could be also an artefact of taxonomic resolution since current study classified down to family level only, while lower taxonomic levels show greater diversity (Guerold, 2000), though significant correlation between species richness and taxon richness at genera, family and class was found among macroinvertebrates (Jiang *et al.*, 2013). Or it could be also due to sampling size as increased sampling size increases diversity (Feeley & Kelly-Quinn, 2011) and richness with increase in number of macroinvertebrates (Graca *et al.*, 2004), and the total sampled area for the whole study area was only 2.7m² with only three samples taken at each reach.

Table 3. Macroinvertebrate diversity of the streams and the whole study area.

Streams	Diversity Indices				
	H'	D'	J'	S	Abundance (no.)
Yuelghona	1.09	.51	.10	11	387
Jichulum	1.07	.07	.07	16	295
Toebirong-chhu	1.71	.28	.16	11	147
Whole study area	1.69	.28	.09	19	829

Table 4. Macroinvertebrate diversity among reaches of the streams.

Streams	Reach No.	Diversity Indices			
		H'	D'	J'	S
Yuelghona	Y1	0.79	0.61	0.44	6.00
	Y2	1.23	0.35	0.69	6.00
	Y3	1.39	0.53	0.87	5.00
	Y4	0.84	0.55	0.47	6.00
Jichulum	J1	0.70	0.72	0.34	8.00
	J2	0.96	0.42	0.70	4.00
	J3	0.73	0.68	0.37	7.00
	J4	1.09	0.45	0.67	5.00
Toebirongchhu	T1	1.72	0.18	0.88	7.00
	T2	1.21	0.47	0.58	8.00

Macroinvertebrate composition

In the whole study area macroinvertebrate community composition at the order level was dominated by Ephemeroptera with 5 families followed by Trichoptera with 3 families (Table 2). The higher numbers of families in Ephemeroptera was also recorded in other studies in other parts of the World (Sharma *et al.*, 2005; Flores & Zafaralla, 2012; Che Salmah *et al.*, 2013b) and it is due to the greater range of habitat occupied by Ephemeroptera when compared to other insect orders, and rocky streams being their centre of abundance and diversity (Thorp & Rogers, 2011). At order level also the orders with fewer families correspond with studies in other parts of the world (Flores & Zafaralla, 2012; Che Salmah *et al.*, 2013b). But Trichoptera dominated in relative abundance while Plecoptera had the least relative abundance, and family level relative abundance also corresponded to the order level relative abundance. The higher relative abundance of Trichoptera could be due to its generalist nature in habitat requirement (Thorp & Rogers, 2011) and also due to pollution sensitivity differences among different families in the study (Giri & Singh, 2012). Hydropsychidae, at the family level, was the most abundant at the whole study area or among the reaches since it is relatively more tolerant to

organic pollution, while order Plecoptera is the least abundant family being very sensitive, and therefore the composition may be the result of anthropogenic activities as Toebirongchhu sub-watershed is critical by function (Wangchuk, 2010), and composition changes predictably in response to stress (Boyle & Fraleigh Jr., 2003; Giri & Singh, 2012; Giri & Singh, 2013). In general the macroinvertebrate composition in the study area reflects the trend of higher diversity of mayflies, stoneflies and caddisflies in higher latitudes (Al-Shami *et al.*, 2013), though stonefly was not dominant in the study area probably due to organic pollution.

Macroinvertebrate diversity and environmental variables

There were no remarkable differences in environmental variables among the reaches except for higher conductivity in the two reaches in Jichulum over other four reaches in Yuelghona and Toebirongchhu streams (Table 5). Among the variables the temperature was significantly correlated with latitude ($\tau=.63$, $P=.018$), longitude ($\tau=.83$, $P=.002$), stream order ($\tau=.63$, $P=.04$), velocity ($\tau=-.75$, $P<.001$) and altitude ($\tau=-.83$, $P=.002$) (Table 6). Also stream order was significantly correlated with width, depth, latitude, longitude ($\tau=.59$, $P=.037$) and altitude ($\tau=-.59$, $P=.037$), and velocity with altitude ($\tau=.61$, $P=.015$) and longitude ($\tau=.61$, $P=.015$). The relationship of temperature with other variables in the study area can be explained by the fact that the reaches within the streams were located at different altitudes, and the velocity was higher in the reaches located at higher altitudes within Yuelghona and Jichulum streams

than in the lower altitude and the higher order Toebirongchhu stream. The stream width had significant correlation with H' ($\tau=.51$, $P=.04$) as was the case in study by Flores & Zafaralla (2012) in Philippines and this relationship could be explained by the fact that stream order increased with the width ($\tau=.596$, $P<.05$) and diversity increases in headwater streams along with the order (Clarke *et al.*, 2008). But other environmental variables considered in the study had no significant correlation with the diversity indices though the direction of the relationship between the environmental factors and the diversity indices were as expected and as observed in other studies. Yet, notably conductivity had the highest correlation with S ($\tau = .65$, $p=.08$) and high correlation with H' and J' ($\tau = -.48$, $p=.19$) as was found in other study in Bhutan (Ofenbock *et al.*, 2010).

Table 5. The physicochemical and hydrological variables within stream reaches in the study area.

Reach	pH	Cond. ($\mu\text{S}/\text{cm}$)	Tem. ($^{\circ}\text{C}$)	Vel. (ms^{-1})	Wid. (cm)	Dep. (cm)
Y1	7.85	62.2	8	1.7	186	27.1
Y2	-	-	8	2.7	185	28.1
Y3	7.67	63.5	8	1.9	223	7.6
Y4	-	-	8	1.9	150	15.5
J1	7.67	154.8	11	1.3	100	21.7
J2	-	-	10	1.5	139	14.3
J3	7.97	150.7	10	1.36	96	10.0
J4	-	-	10	1.5	132	18.8
T1	7.7	83.7	11	1.6	559	70.0
T2	7.83	90.4	11.6	1.0	810	75.0

- Measurements not taken.

Table 6. Correlation (Kendall's tau) among environmental variables.

		pH	Con.	Tem.	Vel.	Wid.	Dep.	Alt.	Lat.	Lon.	Ord.
pH	τ	1.00	.000	.000	-.138	-.138	.138	.000	.000	.000	.000
	p	.	1.00	1.000	.702	.702	.702	1.00	1.00	1.00	1.00
Conductivity	τ	.000	1.00	.501	-.600	-.067	.067	-.200	.467	.200	.000
	p	1.00	.	.173	.091	.851	.851	.573	.188	.573	1.00
Temperature	τ	.000	.501	1.000	-.748**	-.025	.378	-.832**	.630*	.832**	.634*
	p	1.00	.173	.	.006	.925	.156	.002	.018	.002	.040
Velocity	τ	-.138	-.600	-.748**	1.000	.205	-.114	.614*	-.477	-.614*	-.305
	p	.702	.091	.006	.	.417	.652	.015	.058	.015	.293
Width	τ	-.138	-.067	-.025	.205	1.00	.422	-.022	.333	.022	.596*
	p	.702	.851	.925	.417	.	.089	.929	.180	.929	.037
Depth	τ	.138	.067	.378	-.114	.422	1.00	-.333	.200	.333	.596*
	p	.702	.851	.156	.652	.089	.	.180	.421	.180	.037
Altitude	τ	.000	-.200	-.832**	.614*	-.022	-.333	1.00	-.600*	-1.000**	-.596*
	p	1.00	.573	.002	.015	.929	.180	.	.016	.	.037
Latitude	τ	.000	.467	.630*	-.477	.333	.200	-.600*	1.00	.600*	.596*
	p	1.00	.188	.018	.058	.180	.421	.016	.	.016	.037
Longitude	τ	.000	.200	.832**	-.614*	.022	.333	-1.00**	.600*	1.00	.596*
	p	1.00	.573	.002	.015	.929	.180	.	.016	.	.037
Stream order	τ	.000	.000	.634*	-.305	.596*	.596*	-.596*	.596*	.596*	1.000
	p	1.00	1.00	.040	.293	.037	.037	.037	.037	.037	.

*. Correlation is significant at the 0.05 level (2-tailed); **. Correlation is significant at the 0.01 level (2-tailed).

Table 7. Correlation (Kendall's tau) among diversity indices, and diversity indices with environmental variables.

		pH	Con.	Tem.	Vel.	Wid.	Dep.	Alt.	Lat.	Lon.	Ord.	Abu.	H'	D'	J'	S
Shannon-Weiner	τ	-.276	-.467	-.025	.296	.511*	.200	.022	.022	-.022	.447	-.378	1.00	-.689**	.778**	-.215
	p	.444	.188	.925	.241	.040	.421	.929	.929	.929	.117	.128	.	.006	.002	.408
Simpson	τ	.138	.333	-.025	-.30	-.378	-.244	.111	.022	-.111	-.373	.511*	-.689**	1.00	-.822**	.310
	p	.702	.348	.925	.241	.128	.325	.655	.929	.655	.192	.040	.006	.	.001	.232
Evenness	τ	-.276	-.467	-.025	.30	.378	.067	-.02	.067	.022	.298	-.422	.778**	-.822**	1.00	-.454
	p	.444	.188	.925	.24	.128	.788	.929	.788	.929	.296	.089	.002	.001	.	.081
Richness	τ	.000	.645	.487	-.42	.119	.454	-.36	.406	.358	.480	.310	-.215	.310	-.454	1.00
	p	1.00	.079	.081	.115	.646	.081	.168	.118	.168	.109	.232	.408	.232	.081	.

*. Correlation is significant at the 0.05 level (2-tailed); **. Correlation is significant at the 0.01 level (2-tailed)

The pH was negatively correlated with H' ($\tau = -.28$, $p = .44$) while in other studies it was positively correlated (Flores & Zafaralla, 2012) or had mixed response (Che Salmah *et al.*, 2013b), but Flores & Zafaralla (2012) mentions pH of 6.5 to 9.0 is suitable for fish and benthic macroinvertebrates, and pH in the study ranged from 7.67 to 7.97 which could explain its statistically insignificant correlation with the diversity indices.

Among the diversity indices S had stronger correlation with environmental variables than other indices (Table 7) showing its importance as diversity indicator in the study area, and it has been used in many diversity studies elsewhere for different organisms (Graca *et al.*, 2004; Choudhury, Choudhury and Agarwala, 2012; Ali, Bhat and Ali, 2012; Karunaratna *et al.*, 2012; Xiang *et al.*, 2013). The macroinvertebrate diversity for the whole study area, the streams and reaches were relatively low, but it could be due to identification to family level, limited sampling area and single sampling season. However, there was difference in macroinvertebrate composition among the streams and reaches, and also clear increase in diversity with increase in stream order with higher order Toebirongchhu stream and its reaches having higher diversity over that of Jichulum and Yuelghona. Though the composition of macroinvertebrate was dominated by pollution sensitive orders Ephemeroptera and Trichoptera, the other pollution sensitive order Plecoptera was the least dominant with only one family and least abundant. This is a concern as already the streams within the study area are assessed as critical by function and this study records for the first time macroinvertebrate diversity within the study area. But there is urgent need to carry out studies of macroinvertebrate diversity in Bhutan with finer taxonomic resolution and within bigger spatial and longer temporal scales.

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